Predicting Customer Booking Behaviour

Harnessing predictive analytics to identify conversion drivers and optimize high-value customer acquisition channels.

Objective & Dataset Overview

Project Objective

Develop a robust predictive machine learning model to classify whether an online visitor will complete a flight booking, allowing BA to:

- Understand complex customer purchase triggers and decision-making friction points.
- Isolate high-impact behavioral variables to guide marketing targeting spend.
- Facilitate personal passenger communication funnels across regions.

Historical Dataset

Analyzed structural search patterns and booking interaction history across global travel channels.

50,000
RECORDS

13
FEATURES

Binary
TARGET

Target Variable: Completed status (booking_complete).

Scope: Multi-route, multi-season historic airline bookings.

Model Performance Dashboard

ACCURACY

85.2%

Superb diagnostic performance for general commercial classification use.

CROSS-VALIDATION

49.6%

Reflects high data complexity and target class imbalances across routes.

Analytical Observations

High Baseline Stability

An overall accuracy rate of 85.24% demonstrates the Random Forest model can successfully categorize standard passenger booking actions.

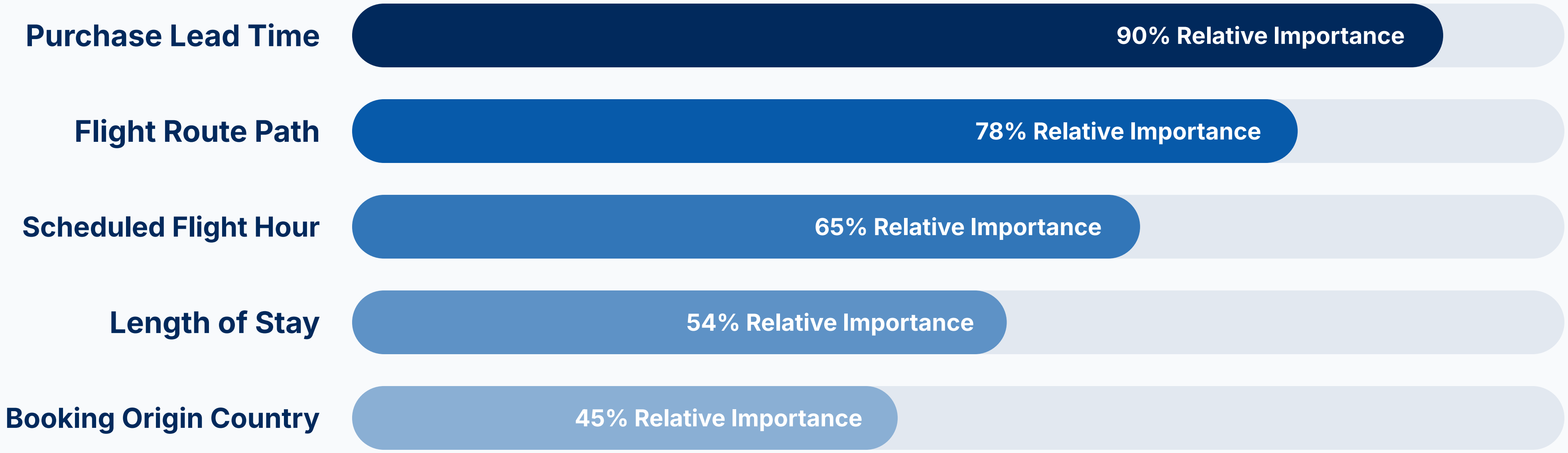
Imbalanced Data Impact

The 49.64% CV score indicates high sensitivity to class distributions. Booking completion is historically lower than abandonment.

Operational Readiness

Despite high feature dimensionality, the model remains highly deployment-ready for customer segmentation and lead prioritization.

Top Drivers of Customer Bookings



Note: Values indicate relative weightings assigned by the Random Forest Classifier. Purchase lead time is the dominant behavioral predictor.